MAINCRAFTS TECHNOLOGY
www.maincrafts.com
hr@maincrafts.com

Artificial Intelligence & Machine Learning- Task 3

Model Validation, Overfitting Control & Hyperparameter Tuning

Objective

In **Task-2**, you learned how to:

- Apply feature scaling and preprocessing
- Train multiple machine learning models
- Compare model performance using RMSE and R^2
- Select the best-performing regression model

Artificial Intelligence & Ma...

In **Task-3**, you will move one step deeper into **professional Machine Learning practice** by learning how to:

- Detect and control overfitting
- Validate models using cross-validation
- Tune model hyperparameters systematically
- Improve generalization performance
- Justify final model selection scientifically

This task focuses on **model reliability**, not just accuracy.

Why This Task Matters (Industry Context)

In real-world ML projects:

- High accuracy on training data is meaningless if the model fails on new data
- Overfitting is one of the most common reasons models fail in production
- Professionals use validation techniques and tuning strategies to build robust models

This task trains interns to:

- Think beyond a single train-test split
- Trust metrics derived from multiple validations
- Build models that generalize well to unseen data

This mirrors **industry-standard ML workflows.**

What You Will Build

You will enhance the **House Price Prediction system** from Task-2 by adding:

- Cross-Validation based evaluation
- Hyperparameter tuning for multiple models
- Overfitting analysis
- Final optimized model selection

The same **California Housing Dataset** will be used to maintain continuity Artificial Intelligence & Ma... .

Concepts You Will Learn

Concept	Purpose
Overfitting & Underfitting	Identify poor generalization
Cross-Validation	Reliable performance estimation
Hyperparameter Tuning	Improve model behavior
GridSearchCV	Automated tuning process
Bias-Variance Tradeoff	Model stability understanding
Model Generalization	Production readiness

Dataset Used

- **California Housing Dataset**
- Target Variable: Median House Value
- Input Features: Same as Task-2 (income, rooms, population, location features)

Dataset source remains unchanged for fair comparison Artificial Intelligence & Ma... .

Tools & Technologies

- Python
- Jupyter Notebook
- pandas, NumPy
- scikit-learn
- matplotlib

Step-by-Step Implementation

Step 1: Import Required Libraries

python

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split, cross_val_score, GridSearchCV
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression, Ridge
from sklearn.tree import DecisionTreeRegressor
from sklearn.metrics import mean_squared_error, r2_score
```

Step 2: Load and Prepare Dataset

```python
data = fetch_california_housing(as_frame=True)
df = pd.concat([data.data, data.target.rename("HousePrice")], axis=1)
```

Separate features and target:

```python
X = df.drop("HousePrice", axis=1)
y = df["HousePrice"]
```

Step 3: Feature Scaling (Same as Task-2)

```python
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

Scaling is required for fair learning and stability, as established in Task-2 Artificial Intelligence & Ma

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Step 4: Train-Test Split

```python
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42
)
```

Step 5: Detect Overfitting (Train vs Test Performance)

Example: Decision Tree

```python
tree = DecisionTreeRegressor(random_state=42)
tree.fit(X_train, y_train)

train_pred = tree.predict(X_train)
test_pred = tree.predict(X_test)

train_rmse = mean_squared_error(y_train, train_pred, squared=False)
test_rmse = mean_squared_error(y_test, test_pred, squared=False)

train_rmse, test_rmse
```

Interpretation

- Large gap between training and test RMSE indicates overfitting
- This is common in tree-based models without constraints

Step 6: Cross-Validation (Reliable Evaluation)

```python
cv_scores = cross_val_score(
    tree, X_scaled, y,
    scoring="neg_root_mean_squared_error",
    cv=5
)

cv_rmse = -cv_scores.mean()
cv_rmse
```

Why Cross-Validation Matters

- Reduces dependency on one train-test split
- Provides stable and realistic performance estimation
- Used by professionals before deployment

Step 7: Hyperparameter Tuning Using GridSearchCV

Example: Decision Tree Optimization

```python
param_grid = {
    "max_depth": [3, 5, 7, 10],
    "min_samples_split": [2, 5, 10]
}

grid = GridSearchCV(
    DecisionTreeRegressor(random_state=42),
    param_grid,
    scoring="neg_root_mean_squared_error",
    cv=5
)

grid.fit(X_train, y_train)
```

Best parameters:

```python
grid.best_params_
```

Step 8: Evaluate Optimized Model

```python
best_tree = grid.best_estimator_

y_pred = best_tree.predict(X_test)

rmse = mean_squared_error(y_test, y_pred, squared=False)
r2 = r2_score(y_test, y_pred)

rmse, r2
```

Compare these results with:

- Baseline Linear Regression
- Ridge Regression (from Task-2)

Step 9: Model Comparison Summary Table

```python
results = {
    "Model": ["Linear Regression", "Ridge Regression", "Tuned Decision Tree"],
    "RMSE": [lr_rmse, ridge_rmse, rmse],
    "R2 Score": [lr_r2, ridge_r2, r2]
}

pd.DataFrame(results)
```

Step 10: Final Model Selection Justification

Intern must explain:

- Why this model was selected
- How overfitting was reduced
- Why cross-validation results are trusted
- Trade-offs between simplicity and performance

This justification is **mandatory**.

Expected Output

You should have:

- Cross-validated performance scores
- Tuned model with controlled overfitting
- Comparison with previous Task-2 models
- Clear explanation of model reliability

Key Learning Outcomes

After completing Task-3, interns will be able to:

- Detect overfitting and underfitting
- Apply cross-validation correctly
- Tune hyperparameters professionally
- Build generalizable ML models
- Follow industry-aligned ML workflows

Deliverables

Mandatory:

1. Jupyter Notebook
   `AI_ML_Task3_Model_Validation_Tuning.ipynb`
2. Cross-validation results
3. Hyperparameter tuning results
4. Model comparison table
5. 2–3 page PDF explaining:
   - Overfitting analysis
   - Tuning approach
   - Final model justification

Optional:

- Additional visualization plots
- Saved trained model using `joblib`

1. Core Tools & Environment (Free)

Python

https://www.python.org/downloads/

Jupyter Notebook

https://jupyter.org/install

Anaconda (Optional, All-in-One)

https://www.anaconda.com/products/distribution

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2. scikit-learn Fundamentals (Must Know)

Official scikit-learn Documentation

https://scikit-learn.org/stable/documentation.html

scikit-learn User Guide (Model Evaluation & Selection)

https://scikit-learn.org/stable/model_selection.html

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3. Overfitting & Underfitting (Conceptual Clarity)

Overfitting vs Underfitting Explained Clearly

https://www.youtube.com/watch?v=Gv9_4yMHFhI

Bias-Variance Tradeoff (Very Important)

https://www.youtube.com/watch?v=EuBBz3bI-aA

Why Models Fail in Production

https://www.youtube.com/watch?v=6g0t3L9PZxE

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4. Train-Test Split & Model Validation

Train-Test Split Explained

https://www.youtube.com/watch?v=0p3QpO8b3xk

Why Single Split Is Not Enough

https://www.youtube.com/watch?v=9Kp3qZzXz7I

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5. Cross-Validation (Core of Task-3)

Cross-Validation Explained Simply

https://www.youtube.com/watch?v=fSytzGwwBVw

K-Fold Cross Validation (Practical)

https://www.youtube.com/watch?v=TIgfjmp-4BA

cross_val_score in scikit-learn

https://www.youtube.com/watch?v=8nKz9JYyF5Y

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6. Hyperparameter Tuning (Critical Section)

What Are Hyperparameters

https://www.youtube.com/watch?v=6g0t3L9PZxE&t=420s

GridSearchCV Explained Step by Step

https://www.youtube.com/watch?v=yy6u5D9zH7g

GridSearchCV vs RandomizedSearchCV

https://www.youtube.com/watch?v=HdlDYng8g9s

7. Decision Tree Overfitting Control

Decision Tree Overfitting Explained

https://www.youtube.com/watch?v=7VeUPuFGJHk

max_depth, min_samples_split Explained

https://www.youtube.com/watch?v=9pH8X6Rz7kY

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8. Regression Metrics (RMSE, R^2)

Regression Evaluation Metrics Explained

https://www.youtube.com/watch?v=Q81RR3yKn30

RMSE vs MAE vs R^2

https://www.youtube.com/watch?v=5XjZ3YJ6X5k

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9. Model Comparison & Selection

How to Compare ML Models Correctly

https://www.youtube.com/watch?v=Z7kM6p2ZxGQ

Choosing the Right Model for Deployment

https://www.youtube.com/watch?v=U7p7GJ8m8Yk

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10. Practical End-to-End Tutorials (Task-Aligned)

House Price Prediction with Model Tuning

https://www.youtube.com/watch?v=J4Wdy0Wc_xQ

Regression Project with scikit-learn

https://www.youtube.com/watch?v=fwY9Qv96DJY

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11. Free Academic & Structured Learning

NPTEL – Machine Learning (Free, High Quality)

https://nptel.ac.in/courses/106/106/106106139

Google Machine Learning Crash Course (Free)

https://developers.google.com/machine-learning/crash-course

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12. Free Reading & Reference Material

Search using these keywords:

- "Cross validation machine learning PDF"
- "GridSearchCV explained PDF"
- "Overfitting and underfitting in ML PDF"
- "Bias variance tradeoff PDF"

Trusted sources:

- https://nptel.ac.in
- https://scikit-learn.org
- https://towardsdatascience.com (free articles)

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13. Practice & Experimentation Resources

scikit-learn Example Gallery

https://scikit-learn.org/stable/auto_examples/index.html

Kaggle Notebooks (Free)
https://www.kaggle.com/notebooks
(Search: “Regression model tuning”, “GridSearchCV regression”)

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14. Community Support (Free Doubt Resolution)

- Stack Overflow (scikit-learn tag)
  https://stackoverflow.com/questions/tagged/scikit-learn
- Reddit Machine Learning
  https://www.reddit.com/r/MachineLearning/
- Reddit Learn ML
  https://www.reddit.com/r/learnmachinelearning/

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15. Recommended Learning Path for Task-3

Day 1

- Overfitting and bias-variance
- Train vs test performance

Day 2

- Cross-validation theory
- cross_val_score practice

Day 3

- Hyperparameters
- GridSearchCV implementation

Day 4

- Tuned model evaluation
- Comparison with baseline models

Day 5

- Documentation
- Final justification write-up

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16. Essentials Checklist for Task-3

You must clearly understand:

- Overfitting and underfitting detection
- Cross-validation purpose and usage
- Hyperparameter tuning logic
- GridSearchCV workflow
- RMSE and R^2 interpretation
- Model selection justification